

**Third Semester B. Tech. (Electronics and Communication Engineering)
Examination****ENGINEERING MATHEMATICS**

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

Use of Non – programmable calculator is permitted.

1. Solve any Two :

(a) Determine whether the following functions are analytic or not :

(i) $f(z) = \bar{z}^2$,

(ii) $f(z) = \sin(\bar{z})$. 6(CO2)

(b) In complex electric field, the potential is $\omega = \phi + i\psi$, where $\psi = x^3 - 3xy^2$, find ϕ . 6(CO2)(c) Find the image of the rectangle whose vertices are (0, 0), (1, 0), (1, 2), (0, 2) by means of the linear transformation $w = (1 + i)z + 2 - i$. Also sketch the image. 6(CO2)**2. Solve any Two :**(a) Using Cauchy's integral formula, evaluate $\int_C \frac{z+1}{(z^3-4z)} dz$, where C is :

(i) $|z+2| = 3/2$,

(ii) $|z-2| = 3/2$. 6(CO2)

(b) Evaluate by contour integration : $\int_0^{2\pi} \frac{d\theta}{2 + \cos \theta}$. 6(CO2)

- (c) Find the Laurent's expansion for $f(z) = \frac{7z - 2}{(z + 1)z(z - 2)}$ in the region given by :

(i) $0 < |z + 1| < 1,$

(ii) $|z + 1| > 3.$ 6(CO2)

3. Solve any **Two** :

- (a) Solve the following Lagrange's linear equation :

$$y^2zp + x^2zq = y^2x \quad \text{where } p = \frac{\partial z}{\partial x} \text{ and } q = \frac{\partial z}{\partial y}. \quad 6(\text{CO1})$$

- (b) Solve : $\frac{\partial^2 z}{\partial x^2} + 3 \frac{\partial^2 z}{\partial x \partial y} + 2 \frac{\partial^2 z}{\partial y^2} = 24xy.$ 6(CO1)

- (c) Using method of separation of variables, solve :

$$3p + 2q = 0, \quad u(x, 0) = 4e^{-x} \quad \text{where } p = \frac{\partial u}{\partial x} \text{ and } q = \frac{\partial u}{\partial y}. \quad 6(\text{CO1})$$

4. Solve :

- (a) Find inverse Laplace Transform of following :

(i) $\frac{2s^2 + 3s + 4}{5s^3}$

(ii) $\frac{1}{2s - 3}.$ 6(CO3)

- (b) Solve by Laplace Transform method :

$$\frac{d^2 y}{dt^2} + 2 \frac{dy}{dt} + 2y = 4e^{-t} t \sin t,$$

given that at $t = 0, y = 0$ and $\frac{dy}{dt} = 0.$ 6(CO3)

5. Solve :

(a) Find Z-transform of following :

(i) $Z\{n^2\}$

(ii) $Z\{na^n\}, a > 0$

6(CO3)

(b) Solve by Z-transform method : $y_{n+2} - 4y_{n+1} + 4y_n = 0$ where $y_0 = 1, y_1 = 0$.
6(CO3)

